

- Count in bases (from 1_{10} to 20_{10})
 - 1) 8
 - 2) 5
 - 3) 3
- Convert from Bin to Dec
 - 1) Convert 1011_2 to decimal.
 - 2) Convert 11010_2 to decimal.
- Convert from Dec to Bin
 - 1) Convert 45_{10} to binary.
 - 2) Convert 128_{10} to binary.
- Convert from Hex to Dec
 - 1) Convert $1A_{16}$ to decimal.
 - 2) Convert $F.F_{16}$ to decimal.
- Convert from Dec to Hex
 - 1) Convert 255_{10} to hexadecimal.
 - 2) Convert 4096_{10} to hexadecimal.
- Convert from Hex to Bin
 - 1) Convert $A3_{16}$ to binary.
 - 2) Convert $9E_{16}$ to binary.
- Convert from Bin to Hex
 - 1) Convert 11010111_2 to hexadecimal.
 - 2) Convert 1011001_2 to hexadecimal.
- Convert from any Base to any Base
 - 1) Convert 212_3 to base 5.
 - 2) Convert 43_7 to base 4.
- Most and Least Significant Digits
 - 1) Identify the MSB and LSB of $(1101001)_2$.
 - 2) Identify the MSD and LSD of $(3A7)_{16}$.
- Floating Point Representation
 - 1) Represent 12.625_{10} in single-precision IEEE 754.
 - 2) Represent -0.15625_{10} in single-precision IEEE 754.
 - 3) Convert the IEEE 754 value $0xC1480000$ to decimal.
 - 4) Convert the IEEE 754 value $0x40A00000$ to decimal.
- Convert from Bin to IEEE 754
 - 1) Represent 101.11_2 in IEEE 754 single precision.
 - 2) Represent -0.101_2 in IEEE 754 single precision.
- Convert from Bin to 2's Complement
 - 1) Find the 8-bit 2's complement of 101101_2 .
 - 2) Find the 8-bit 2's complement of -15_{10} .
- Convert from any Base to IEEE 754
 - 1) Convert $(12.5)_{10}$ to IEEE 754 single precision.
 - 2) Convert $(A.8)_{16}$ to IEEE 754 single precision.
- Convert from any Base to 2's Complement
 - 1) Represent -20_{10} in 8-bit 2's complement.
 - 2) Represent -12_8 in 8-bit 2's complement.